# Homework 9

Due November 25, 2025.

## Problem 1.

Evaluate the Clebsch-Gordan (CG) decomposition of the product of two adjoint matrices of $SU(3)$. Show that there are two adjoint matrices in this decomposition. Using that CG coefficients are invariant tensors, explain what the relation is between the two adjoint matrices in the decomposition, and the structure constants and the $d$-symbol. Could there be 3 adjoint matrices in the CG expansion of two adjoint matrices of $SU(N)$? One adjoint matrix? Show that for $SU(N)$ there are again always two adjoint matrices in the expansion.

## Problem 2.

Writ down the CG decomposition of the product of two spinor irreps of $SO(10)$, and of the product of one spinor irrep and one conjugate spinor irrep. First construct the charge conjugation matrix in $d = 10$, and determine if it is block-diagonal or block-off-diagonal. Are the spinor and conjugate spinor irreps in $d = 10$ real, pseudoreal, or complex?